MIL-T-60116(MU)
17 June 1964
SUPERSEDING
FA-PD-MI-2350, Rev. 3
23 November 1962

MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED, M20A1
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated, M20A1, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military
- MIL-A-2550 - Ammunition and Special Weapons, General Specification for
- MIL-I-45208 - Inspection System Requirements.

STANDARDS

Military
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions
- MIL-STD-413 - Visual Inspection Guide for Rubber O-Rings
- MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

Munitions Command
- F10533858 - Thruster, Cartridge Actuated, M20A1, Metal Parts Assembly
- C8797741 - Container, Ammunition, Fiber M347 for Thruster, Cartridge Actuated, M20A1 or M5A3

FSC 1375

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DRAWINGS, Cont.

C8797742 - Box, Packing, Ammunition for Thruster, Cartridge Actuated M20A1 or M5A3 in Fiber Container M347.
IEL-10533858- Inspection Equipment List for Thruster, Cartridge Actuated M20A1

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General - Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Initial production sample - Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Material - The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Physical properties - Physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.5 Hydrostatic pressure - The body and head shall withstand a constant internal pressure of 8000 pounds per square inch (psi) minimum, for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1).

3.6 Structural strength - The assembled thruster shall withstand a static tensile force of 2000 pounds, minimum for 15 seconds, minimum, between mounting points without unlocking and without mechanical failure (see 6.1).

3.7 Metal parts assembly - Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing.

3.7.1 The head and body assembly and the retainer guide and sleeve assembly shall be assembled hand tight.

3.7.2 Trunnion rotation - Upon completion of assembly, the trunnion shall be free to rotate 360 degrees.

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3.7.3 End sleeve rotation - Upon completion of assembly and after structural strength test, the end sleeve shall be free to rotate 360 degrees in a clockwise direction.

3.8 Protective finish - Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.9 Residual magnetism - The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.10 Workmanship - The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.10.1 Metal defects - All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.10.2 Burr - No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.3 Foreign matter - No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions - Reference is made to Standard MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection - Inspection shall be in accordance with Specifications MIL-A-2550 and MIL-I-45208, except as specified herein.

4.2 Initial production sample.- At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 5 metal parts assemblies, 10 sets of parts and fifty shear pins from the same lot of shear pins tested by the contractor. The samples shall be subjected to the tests in

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4.3.3. Test procedures shall be in accordance with 4.4, except as provided in Specification MIL-I-45208. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as used in regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of Specification MIL-I-45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions

4.3.1 Lot

4.3.1.1 Submission of product- The product shall be submitted in accordance with Standard MIL-STD-105.

4.3.1.2 Size of lot - The shipping lot of metal parts assemblies shall consist of 318 thrusters, maximum, provided the homogeneity requirements of inspection lots have been retained. A small lot may be shipped upon approval of the contracting officer.

4.3.2 Examination - One hundred per cent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and associated criteria in table I and II. For an individual defect (single characteristic) the criteria in table I shall be used. For total defects within a major or minor group the criteria in table II shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in Standard MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber O-rings shall be in accordance with Standard MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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Table I

Lot Size	Sample Size	Major Accept	Major Reject	Minor Accept	Minor Reject
1 - 300	100% Examination				
301 - 1,300	110	1	2	2	3
1,301 - 3,200	150	2	3	3	4
3,201 - 8,000	225	3	4	4	5
8,001 - 22,000	300	4	5	5	6

Table II

Total Allowable Defects Per Sample Size

Sample Sized for Major Defects	110	150	225	300
No. of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14

Sample Sizes for Minor Defects	110	150	225	300
No. of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Key, Drawing C8593857 a detail of Drawing F10533858.

Defect	Method of Inspection
CRITICAL	
None defined	

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Defect		Method of Inspection
MAJOR		
101.	Radial profile of length, inside and outside	Gage
102.	Radial profile of width	Gage
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Surface finish, improper	Visual
106.	Burr or sharp corner	Visual
107.	Foreign matter	Visual
MINOR		
201.	Total length	Gage
202.	Radius missing or incorrect, except as otherwise classified	Visual

4.3.2.2 Pin, Firing Drawing D8593864 a detail of Drawing F10533858.

CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of groove with body	Gage
102.	Concentricity of striker with body	Gage
103.	Concentricity of striker shoulder with body	Gage
104.	True position of hole	Gage
105.	Diameter of groove	Gage
106.	Diameter of body, min.	Gage
107.	Diameter of striker, max.	Gage
108.	Diameter of hole, max.	Gage
109.	Profile of striker	Gage
110.	Profile of groove	Measure
111.	Metal defective	Visual
112.	Protective finish inadequate or defective	Visual
113.	Surface finish of body and striker tip, improper	Visual
114.	Burr or sharp corner	Visual
115.	Foreign matter	Visual

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Defect		Method of Inspection
MINOR		
201.	Diameter of body, max.	Gage
202.	Diameter of striker, min.	Gage
203.	Diameter of hole, min.	Gage
204.	Diameter of striker shoulder	Gage
205.	Depth of hole	Gage
206.	Length from body to tip of striker	Gage
207.	Length of striker	Gage
208.	Total length	Gage
209.	Location of hole	Gage
210.	Location of groove	Gage
211.	Depth of slot in base	Measure
212.	Length of body diameter forward of groove, min.	Measure
213.	Width of groove	Measure
214.	Width of slot in base	Measure
215.	Location of slot in base	Visual
216.	Marking missing, incorrect or illegible	Visual
217.	Radius missing or incorrect	Visual
218.	Surface finish improper, except as otherwise classified	Visual

4.3.2.3 Body Drawing D8594480 a detail of Drawing F10533858.

CRITICAL		
None defined		
MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of pitch diameter of thread with cavity	Gage
104.	Concentricity of bore with cavity	Gage
105.	Concentricity of groove with bore	Gage
106.	Concentricity of counterbore with bore	Gage
107.	Diameter of cavity, max.	Gage
108.	Diameter of thread undercut, min.	Gage
109.	Diameter of counterbore	Gage
110.	Diameter of groove	Gage
111.	Diameter of bore	Gage
112.	Diameter of body, thread end	Gage
113.	Diameter of internal undercut, max. if applicable	Gage
114.	Location of groove	Gage
115.	Straightness of bore	Gage

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MAJOR (Cont'd

	Defect	Method of Inspection
116.	Profile of groove	Measure
117.	Thread damaged or not full	Visual
118.	Metal defective	Visual
119.	Protective finish inadequate or defective	Visual
120.	Surface finish of cavity, improper	Visual
121.	Burr or sharp corner	Visual
122.	Foreign matter	Visual

MINOR

201.	Pitch diameter of thread, max.	Gage
202.	Diameter of cavity, min.	Gage
203.	Diameter of body, max.	Gage
204.	Depth of counterbore	Gage
205.	Total length	Gage
206.	Length to exterior shoulder	Gage
207.	Width across flats	Gage
208.	Thickness of wall	Gage
209.	Diameter of shoulder	Measure
210.	Length of base of cavity	Measure
211.	Length of shoulder	Measure
212.	Width of thread undercut, max.	Measure
213.	Width of internal undercut, max., if applicable	Measure
214.	Width of flat	Measure
215.	Location of thread undercut	Measure
216.	Location of flats, incorrect	Visual
217.	Marking missing, incorrect or illegible	Visual
218.	Radius or chamfer, missing or incorrect	Visual
219.	Surface finish improper, except as otherwise classified	Visual

4.3.2.4 Trunnion, Drawing D8594482 a detail of Drawing F10533858.

CRITICAL

None defined

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Defect	Method of Inspection
MAJOR	
101. Concentricity of lugs	Gage
102. True position of lugs	Gage
103. Symmetry of flats at root of lugs with inside	Gage
104. Perpendicularity of sides with inside	Gage
105. Diameter of lugs, min.	Gage
106. Width across flats, min.	Gage
107. Radius at base of lugs	Gage
108. Metal defective	Visual
109. Protective finish inadequate or defective	Visual
MINOR	
201. Diameter of lugs, max.	Gage
202. Diameter, outside, max.	Gage
203. Diameter, inside	Gage
204. Length of lugs	Gage
205. Width across flats, max.	Gage
206. Width of body	Gage
207. Thickness of wall	Gage
208. Location of lugs	Gage
209. Marking missing incorrect or illegible	Visual
210. Radius or chamfer missing or incorrect, except as otherwise classified	Visual
211. Surface finish, improper	Visual
212. Burr or sharp corner	Visual
213. Foreign matter	Visual

4.3.2.5 Screw, Buffer, Drawing D8594485 a detail of Drawing F10533858.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of flange, groove end, with thread	Gage
104. Diameter of groove	Gage
105. Diameter of flange, groove end, min.	Gage
106. Diameter of head, min.	Gage
107. Width of groove	Gage

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Defect	Method of Inspection
MAJOR (Cont'd)	
108. Profile of groove	Measure
109. Thread damaged or not full	Visual
110. Metal defective	Visual
111. Protective finish inadequate or defective	Visual
112. Burr or sharp corner	Visual
113. Foreign matter	Visual
MINOR	
201. Pitchdiamater of thread, max.	Gage
202. Diameter of head, max.	Gage
203. Diameter of thread undercut, min.	Gage
204. Diameter of body	Gage
205. Diameter of flange, groove end, max.	Gage
206. Diameter of flange, thread end	Gage
207. Length of threaded end	Gage
208. Length of body from thread flange to head	Gage
209. Total length	Gage
210. Thickness of flange, groove end	Gage
211. Width of undercut	Measure
212. Width of flange, thread end	Visual
213. Slot incorrect	Visual
214. Marking missing, incorrect or illegible	Visual
215. Radius or chamfer missing or incorrect	Visual
216. Surface finish, improper	Visual

4.3.2.6 Sleeve, End Drawing D8596446 a detail of Drawing F10533858.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of shoulder with body	Gage
104. Concentricity of large cavity with body	Gage
105. Concentricity of bore with large cavity	Gage
106. Concentricity of small cavity with large cavity	Gage
107. Symmetry of slots with body	Gage
108. Diameter of body, min.	Gage

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Defect	Method of Inspection
MAJOR (Cont'd)	
109. Diameter of large cavity, max.	Gage
110. Diameter of small cavity, max.	Gage
111. Diameter of bore	Gage
112. Diameter of shoulder, min.	Gage
113. Diameter of pin hole	Gage
114. Location of pin hole	Gage
115. Angel of pin hole	Gage
116. Width of slots	Gage
117. Location of slots	Gage
118. Radius at thread end	Measure
119. Thread damaged or not full	Visual
120. Metal defective	Visual
121. Protective finish inadequate or defective	Visual
122. Burr or sharp corner	Visual
123. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of body, max.	Gage
203. Diameter of large cavity, min.	Gage
204. Diameter of small cavity, min.	Gage
205. Diameter of shoulder, max.	Gage
206. Diameter of flange	Gage
207. Diameter of thread undercut, min.	Gage
208. Length of small cavity at midsection	Gage
209. Length of slots	Gage
210. Variation in alignment, flange face with face of large cavity seat	Gage
211. Depth of large cavity	Measure
212. Total length	Measure
213. Width of shoulder	Measure
214. Width of thread undercut	Measure
215. Width across flats	Measure
216. Width of flange at flat	Measure
217. Location of undercut	Measure
218. Location of flat	Measure
219. Radius or chamfer missing or incorrect, except as otherwise classified	Visual
220. Marking missing, incorrect or illegible	Visual
221. Surface finish, improper	Visual

4.3.2.7 Spring, Piston Locking, Drawing C8594508 a detail of Drawing F10533858.

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Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Load at specified height	Gage
102. Solid length max.	Gage
103. Ends incorrect	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
None defined	

4.3.2.8 Head Drawing D10521455 a detail of Drawing F10533858.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of small thread, max.	Gage
102. Minor diameter of small thread, max.	Gage
103. Pitch diameter of midsection thread, max.	Gage
104. Minor diameter of midsection thread, max.	Gage
105. Pitch diameter of large thread, max.	Gage
106. Minor diameter of large thread, max.	Gage
107. Concentricity of pitch diameter of large thread with pitch diameter of midsection thread	Gage
108. Depth of cavity, small end	Gage
109. Depth of counterbore, small end	Gage
110. Length of thread, small end, max.	Gage
111. Length of effective thread, small end, min.	Gage
112. Diameter of undercut, large thread, max.	Measure
113. Diameter of undercut, midsection thread, max.	Measure
114. Diameter of chamfer at small end	Measure
115. Angle of chamfer at small end	Measure
116. Thread damaged or not full	Visual
117. Metal defective	Visual
118. Protective finish inadequate or defective	Visual
119. Surface finish of face in midsection cavity, chamfers in small end and small thread face, improper	Visual

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Defect	Method of Inspection
MAJOR (Cont'd)	
120. Burr	Visual
121. Foreign matter	Visual
MINOR	
201. Pitch diameter of small thread, min.	Gage
202. Pitch diameter of midsection thread, min.	Gage
203. Pitch diameter of large thread, min.	Gage
204. Diameter of counterbore, small end	Gage
205. Diameter of hole thru midsection	Gage
206. Depth of large cavity	Gage
207. Diameter of body, midsection	Measure
208. Diameter of body, large end	Measure
209. Diameter of body, small end	Measure
210. Depth to bottom of midsection	Measure
211. Length of large body diameter	Measure
212. Length of small body diameter	Measure
213. Length of flat	Measure
214. Total length	Measure
215. Width of undercut, large thread	Measure
216. Width of undercut, midsection thread	Measure
217. Width across flats	Measure
218. Location of flats	Measure
219. Marking missing, incorrect or illegible	Visual
220. Radius or chamfer missing or incorrect, except as otherwise classified	Visual
221. Surface finish improper, except as otherwise classified	Visual

4.3.2.9 Piston, Drawing F10521456 a detail of Drawing F10533858.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of body midsection with body, large end	Gage
104. Concentricity of body, small end with large end	Gage

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Defect	Method of Inspection
MAJOR (Cont'd)	
105. Concentricity of ring groove with body, large end	Gage
106. Concentricity of pitch diameter of thread with small end of body	Gage
107. Concentricity of small cavity with small end of body	Gage
108. Concentricity of key escape groove with small end of body	Gage
109. Concentricity of locking groove with small end of body	Gage
110. Diameter of body, large end, min.	Gage
111. Diameter of ring groove	Gage
112. Diameter of body, midsection	Gage
113. Diameter of body, small end	Gage
114. Diameter of key escape groove, max.	Gage
115. Diameter of locking groove, min.	Gage
116. Length of thread, max.	Gage
117. Length of effective thread, min.	Gage
118. Location of angle in key groove	Measure
119. Length to end of key escape groove	Measure
120. Length of small end of body, min.	Measure
121. Profile of key escape and locking grooves	Measure
122. Profile of ring groove	Measure
123. Thread damaged or not full	Visual
124. Metal defective	Visual
125. Protective finish inadequate or defective	Visual
126. Surface finish, improper	Visual
127. Burr or sharp corner	Visual
128. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, min.	Gage
202. Diameter of body, large end, max.	Gage
203. Diameter of shoulder, large end	Gage
204. Diameter of key escape groove, min.	Gage
205. Diameter of locking groove, max.	Gage
206. Diameter of undercut, min., if applicable	Gage
207. Diameter of small cavity	Gage
208. Depth of small cavity	Gage
209. Depth of thread, cavity, max.	Gage
210. Length to body, large end	Gage

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Defect	Method of Inspection
MINOR (Cont'd)	
211. Length of shoulder, large end	Gage
212. Length to locking groove	Gage
213. Diameter of large cavity	Measure
214. Depth of large cavity and drill point, max.	Measure
215. Length to body, midsection, max.	Measure
216. Total length	Measure
217. Width of ring groove	Measure
218. Width of undercut, max., if applicable	Measure
219. Location of ring groove	Measure
220. Marking missing, incorrect or illegible	Visual
221. Radius or chamfer missing or incorrect	Visual

4.3.2.10 Retainer, Guide, Drawing D10521615 covering a detail of Drawing F10533858.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of external thread, min.	Gage
102. Major diameter of external thread, min.	Gage
103. Pitch diameter of internal thread, max.	Gage
104. Minor diameter of internal thread, max.	Gage
105. Concentricity of counterbore with pitch diameter of internal thread	Gage
106. Concentricity of boss with pitch diameter of internal thread	Gage
107. Concentricity of body with pitch diameter of external thread	Gage
108. Concentricity of small cavity with large cavity	Gage
109. Concentricity of ring groove with body	Gage
110. Concentricity of hole thru boss with large cavity	Gage
111. Concentricity of large cavity with pitch diameter	Gage
112. Concentricity of small hole thru boss with internal thread and large cavity	Gage
113. Perpendicularity of boss face with pitch diameter of internal thread	Gage
114. True position of pin hole	Gage
115. Diameter of ring groove	Gage

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Defect		Method of Inspection
MAJOR (Cont'd)		
116.	Diameter of large cavity	Gage
117.	Diameter of counterbore, max.	Gage
118.	Diameter of hole thru boss	Gage
119.	Diameter of body	Gage
120.	Diameter of undercut, max.	Gage
121.	Height of boss	Gage
122.	Thickness thru boss	Gage
123.	Thickness of base of large cavity	Gage
124.	Profile of ring groove	Measure
125.	Thread damaged or not full	Visual
126.	Metal defective	Visual
127.	Protective finish inadequate or defective	Visual
128.	Surface finish in large cavity, improper	Visual
129.	Burr or sharp corner	Visual
130.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of external thread, max.	Gage
202.	Pitch diameter of internal thread, min.	Gage
203.	Diameter of small cavity	Gage
204.	Diameter of counterbore, min.	Gage
205.	Diameter at rear of external thread	Gage
206.	Symmetry of flats with pitch diameter of internal thread	Gage
207.	Depth of cavity	Gage
208.	Location of undercut	Gage
209.	Length of effective external thread, min.	Gage
210.	Location of flange	Gage
211.	Total length	Gage
212.	Width across flats	Gage
213.	Location of pin hole	Gage
214.	Location of ring groove	Gage
215.	Length of flats	Measure
216.	Width of ring groove	Measure
217.	Width of undercut, internal thread	Measure
218.	Marking missing, incorrect or illegible	Visual
219.	Radius or chamfer missing or incorrect	Visual
220.	Surface finish, improper, except as otherwise classified	Visual

4.3.2.11 Sleeve, Drawing C10521617 covering a detail of Drawing F10533858.

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Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of bore with pitch diameter of thread	Gage
104. Symmetry of flats with pitch diameter of thread	Gage
105. Diameter of bore	Gage
106. Radius at end of bore, thread end	Gage
107. Thread damaged or not full	Visual
108. Metal defective	Visual
109. Protective finish inadequate or defective	Visual
110. Surface finish on radius at end of bore, thread end, improper	Visual
111. Burr	Visual
112. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of body	Gage
203. Width across flats	Gage
204. Total length	Measure
205. Length of thread	Measure
206. Length of flats	Measure
207. Marking missing, incorrect or illegible	Visual
208. Radius or chamfer missing or incorrect, except as otherwise classified	Visual
209. Surface finish improper, except as otherwise classified	Visual

4.3.2.12 Pin, Shear, Drawing C10521618 covering a detail of Drawing F10533858.

Defect	Method of Inspection
CRITICAL	
None defined	

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Defect	Method of Inspection
MAJOR	
101. Diameter, max.	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Total length	Measure
202. Profile of end	Visual

4.3.2.13 Body and Pellet Assembly, Drawing D8594481 a detail of Drawing F10533858.

CRITICAL	
None defined	
MAJOR	
101. Protrusion of pellets, max.	Gage
102. Pellets loose, missing or improperly inserted	Visual
103. Thread damaged	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Location of pellets	Measure

4.3.2.14 Guide Retainer and Pellet Assembly, Drawing D10521616 a detail of Drawing F10533858.

CRITICAL	
None defined	
MAJOR	
101. Protrusion of pellets, max.	Gage
102. Pellets loose, missing or improperly inserted	Visual
103. Thread damaged	Visual
104. Burr	Visual
105. Foreign matter	Visual

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Defect	Method of Inspection
MINOR	
201. Location of pellets	Measure
4.3.2.15 Buffer Screw and Pellet Assembly, Drawing C8594486 a detail of F10533858.	
CRITICAL	
None defined	
MAJOR	
101. Protrusion of pellets, max.	Gage
102. Pellet Loose, missing or improperly inserted	Visual
103. Thread damaged	Visual
104. Burr.	Visual
105. Foreign matter	Visual
MINOR	
201. Location of pellets	Measure
4.3.2.16 Assembly, Drawing F10533858, less all component parts except, Guide Retainer 10521616, Firing Pin 8593864, O-Ring 501220.	
CRITICAL	
1. Protrusion of the firing pin	Gage
MAJOR	
101. Any part missing	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

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4.3.2.17 Metal Parts Assembly, Drawing F10533858.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Trunnion not free to rotate, 360 degrees	Manual
102. End sleeve not free to rotate, 360 degrees clockwise	Manual
103. Any part missing, except as otherwise classified	Visual
104. Improper assembly	Visual
105. Thread damaged or not full	Visual
106. Marking missing, incorrect or illegible	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
201. Shipping cap or plug missing	Visual
202. Protective finish inadequate or defective	Visual

4.3.2.18 Container, Fiber Drawing C8797741.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damage through all impregnated layers	Visual

4.3.2.19 Box, Packing Drawing C8797742.

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Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual/Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual

4.3.3 Inspection testing - The following tests shall be performed in accordance with the provisions of 4.4 except as provided in Specification MIL-I-45208.

4.3.3.1 Hardness - Fifty units from each heat treated lot of components shall be tested for compliance with the values noted on the applicable drawing. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot of components.

4.3.3.2 Hydrostatic pressure - Each body and each head shall be subjected to an internal hydrostatic pressure. Failure of any unit to comply with the requirements of 3.5 shall be cause for rejection of the unit.

4.3.3.3 Shear pin strength - Fifty pins from each lot shall be tested for shear strength. Failure of any shear pin to comply with the requirements of the drawing shall be cause for rejection of the lot.

4.3.3.4 Structural strength - Each thruster shall be subjected to the structural strength test. Failure of any thruster to comply with the requirements of 3.6 shall be cause for rejection of the thruster.

4.3.3.5 Protective finish - The number of specimens and acceptance criteria protective finish tests shall be in accordance with the applicable specifications listed on the drawings.

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4.3.3.6 Residual magnetism - Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.9 shall be cause for rejection of the assembly.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL 10533858, except as provided in Specification MIL-I-45208.

4.4 Test methods and procedures.

4.4.1 Hardness - Prior to the application of protective finish, a hardness testing machine shall be used to determine the hardness values specified on the detail drawings.

4.4.2 Hydrostatis pressure - Prior to application of protective finish, the body and head, individually, shall be mounted in an approved device. The testing device used shall internally seal the body and head, as applicable, at each end without applying any axial load other than that applied by the liquid pressure.

4.4.3 Shear pin strength - After application of protective finish, the load shall be applied at the steady rate of 0.125 inches per minute. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.4.4 Structural strength - The thruster shall be subjected to the structural strength test in accordance with the requirements of 3.6. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.4.5 Protective finish - Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.4.6 Residual magnetism - The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from and at the same height as an approved compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading as shown in Figure 1. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A - Packing and marking shall be in accordance with Drawings C8797741 and C8797742.

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5.2 Data cards - Data card information shall be as specified in Specification MIL-A-2550.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

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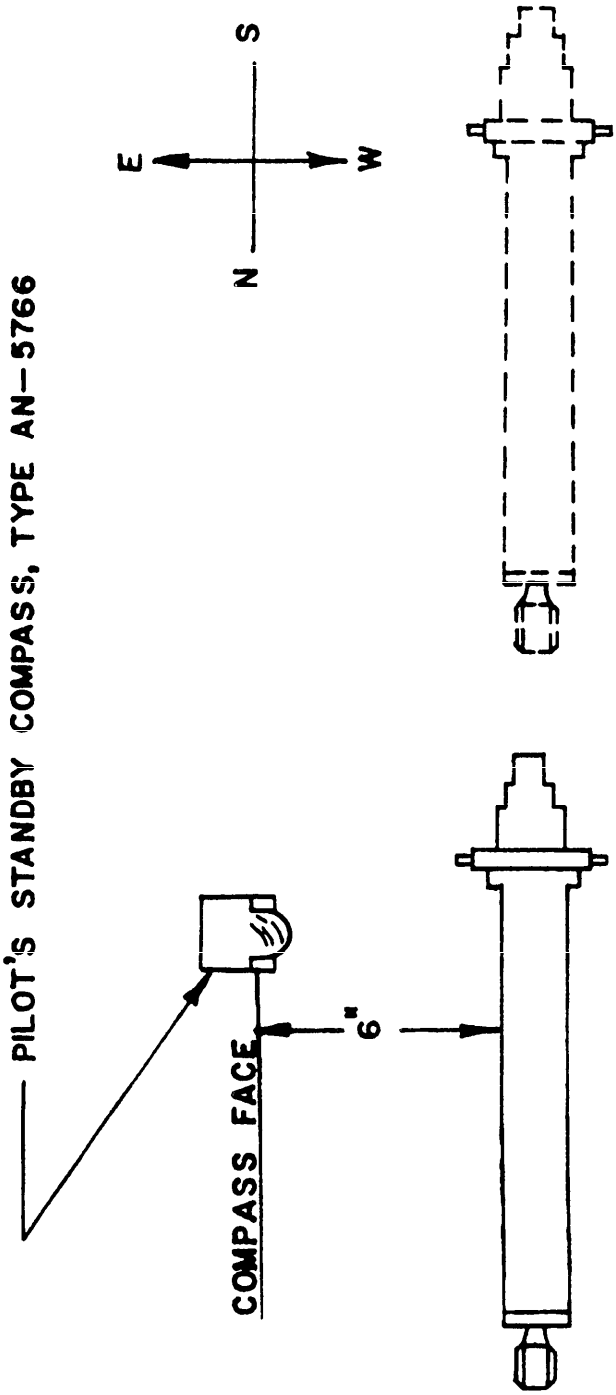


FIGURE 1. LAYOUT FOR DETERMINATION FOR RESIDUAL MAGNETISM IN THRUSTER, CARTRIDGE ACTUATED, M20A1.

GPO 8

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 62 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE

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